

# **MUST lecture 10**

**Two-population classification based on training data  
with known population membership**

## **Disclaimer:**

**Please note that these slides are only a brief overview of the topic.  
Many more details will be explained on the blackboard and with  
MATLAB examples**

# Two-population classification based on training data

**Problem**      **Classify new multivariate observation** as being from **population 1 or 2** based on training data with known population membership

## Training data

- already collected “ground truth” data with known/defined population membership

## New data

- “real” new observations for which definitive establishment of population membership would be
  - too expensive
  - too slow
  - impossible
  - only possible by destructive test

## Applications

- Pattern recognition / detection
- Machine learning
- Digital communication / detection of signals
- Medicine / pathology
- Feature extraction

## Two-population classification: methodology (for only 2 populations $g = 2$ below)

### Model

- $g \geq 2$  populations:  $\pi_1, \pi_2, \dots, \pi_g$
  - $\pi_i$ :  $f_i(\mathbf{x}), \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, i = 1, \dots, g$
  - MVN assumption:  $\mathbf{X}|\pi_i \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i), i = 1, \dots, g$
- 

### Sample space

- $p$ -dimensional multivariate observations,  $\mathbf{x} = [x_1, \dots, x_p]^T \in R^p$
  - training data available for each population  $\rightarrow \hat{f}_i(\mathbf{x}), \hat{\boldsymbol{\mu}}_i, \hat{\boldsymbol{\Sigma}}_i, i = 1, \dots, g$
- 

### Classification

- splitting of sample space  $R^p = \bigcup_{i=1}^g R_i, R_i \cap R_k = \emptyset, i \neq k$
- decision rule for new observations  $d: \mathbf{x}_0 \in R^p \rightarrow d \in \{d_1, d_2, \dots, d_g\}$   
where  $d(\mathbf{x}_0) = d_i$  for  $\mathbf{x}_0 \in R_i$

## Two-population classification: informations used for decision making (for only 2 populations $g = 2$ )

### Misclassifications: Confusion matrix

| Classify as → | $d_1$    | $d_2$    | ... | $d_g$    |
|---------------|----------|----------|-----|----------|
| True $\pi_1$  | $P(1 1)$ | $P(2 1)$ | ... | $P(g 1)$ |
| ↓ $\pi_2$     | $P(1 2)$ | $P(2 2)$ | ... | $P(g 2)$ |
| ...           | ...      | ...      | ... | ...      |
| $\pi_g$       | $P(1 g)$ | $P(2 g)$ | ... | $P(g g)$ |

where  $P(k|i) = \int_{R_k} f_i(\mathbf{x}) d\mathbf{x}, \quad \mathbf{x} \in R^p$

### Misclassifications: Cost matrix

| Classify as → | $d_1$    | $d_2$    | ... | $d_g$    |
|---------------|----------|----------|-----|----------|
| True $\pi_1$  | 0        | $c(2 1)$ | ... | $c(g 1)$ |
| ↓ $\pi_2$     | $c(1 2)$ | 0        | ... | $c(g 2)$ |
| ...           | ...      | ...      | ... | ...      |
| $\pi_g$       | $c(1 g)$ | $c(2 g)$ | ... | 0        |

where  $c(k|i) \geq 0, \quad i \neq k$

### Priors

- $p_i = P(\pi_i), \quad i = 1, \dots, g$   
where  $\sum_{i=1}^g p_i = 1$

## General classification criteria (distribution-free / non-parametric)

### ECM, Expected Cost of Misclassifications

$$ECM = p_1 P(2|1)C(2|1) + p_2 P(1|2)C(1|2) \quad \text{Minimizing gives}$$

$$\frac{f_1(x_0)}{f_2(x_0)} > \frac{p_2 C(1|2)}{p_1 C(2|1)}$$

Special cases

Equal costs

$$\frac{f_1(x_0)}{f_2(x_0)} > \frac{p_2}{p_1}$$

Equal costs and priors

$$\frac{f_1(x_0)}{f_2(x_0)} > 1$$

### MAP criteria, Maximum A posteriori Probability (usefull for soft decisions)

Posteriors  $P(\pi_1|x_0) = \frac{p_1 f_1(x_0)}{P(x_0)}, \quad P(\pi_2|x_0) = \frac{p_2 f_2(x_0)}{P(x_0)}, \quad P(\pi_1|x_0) + P(\pi_2|x_0) = 1$

$$P(\pi_1|x_0) > P(\pi_2|x_0) \Leftrightarrow p_1 f_1(x_0) > p_2 f_2(x_0)$$

## Classification for 2 Multivariate Normal (MVN) distributions

Model  $X|\pi_i \sim N_p(\mu_i, \Sigma_i), \quad i = 1, 2$

Case 1:  $\Sigma_1 = \Sigma_2 \stackrel{\text{def}}{=} \Sigma$

Put MVN pdf's in ECM and simplify giving  
LDA, Linear Discriminant Analysis decision rule

$$(\mu_1 - \mu_2)^T \Sigma^{-1} x_0 - \frac{1}{2} (\mu_1 - \mu_2)^T \Sigma^{-1} (\mu_1 + \mu_2) \begin{matrix} d_1 \\ > \\ d_2 \end{matrix} \log \frac{p_2 c(1|2)}{p_1 c(2|1)}$$

Sample based (training data):

Put in estimates  $\hat{\mu}_1 = \bar{X}_1, \quad \hat{\mu}_2 = \bar{X}_2, \quad \hat{\Sigma} = S_p$

Case 2:  $\Sigma_1 \neq \Sigma_2$

Put MVN pdf's in ECM and simplify giving  
QDA, Quadratic Discriminant Analysis decision rule

$$-\frac{1}{2} x_0^T (\Sigma_1^{-1} - \Sigma_2^{-1}) x_0 + (\mu_1^T \Sigma_1^{-1} - \mu_2^T \Sigma_2^{-1}) x_0 - k \begin{matrix} d_1 \\ > \\ d_2 \end{matrix} \log \frac{p_2 c(1|2)}{p_1 c(2|1)}$$

where  $k = \frac{1}{2} \log \frac{|\Sigma_1|}{|\Sigma_2|} + \frac{1}{2} (\mu_1^T \Sigma_1^{-1} \mu_1 - \mu_2^T \Sigma_2^{-1} \mu_2)$

Sample based (training data):

Put in estimates  $\hat{\mu}_1 = \bar{X}_1, \quad \hat{\mu}_2 = \bar{X}_2, \quad \hat{\Sigma}_1 = S_1, \quad \hat{\Sigma}_2 = S_2$

## Classification performance evaluation

### APER, Apparant Error Rate

Make confusion matrix with counts of correct and incorrect classification of training data

$$APER = \frac{\# \text{ of misclassifications}}{\text{total \# of classifications}}$$

APER is **biased** and underestimates the true error rate because same data (training data) are used first to make decision rule and the again to assess classification succes

### Estimated Actual Error Rate

Use cross-validation or hold-out procedure for one training data observation at a time

Make confusion matrix with counts of correct and incorrect classification

$$\widehat{AER} = \frac{\# \text{ of misclassifications}}{\text{total \# of classifications}}$$

The estimated Actual Error Rate is an **unbiased** estimate of the actual error rate

# Summarizing two-population discriminant analysis procedures

## Training data

- collect “ground truth” data
- define or estimate prior probabilities,  $p_i = P(\pi_i)$ ,  $i = 1, 2$
- define misclassification costs,  $C(1|2)$ ,  $C(2|1)$
- estimate parameters for  $f(\mathbf{x}|\pi_i)$ , i.e.  $\hat{\mu}_i, \hat{\Sigma}_i$ ,  $i = 1, 2$
- MVN model control
- test  $H_0: \Sigma_1 = \Sigma_2 \gg \gg$  LDA case for accept / QDA case for reject
- test  $H_0: \mu_1 = \mu_2 \gg \gg$  significant separation of populations for reject
- specify/calculate sample based discriminant functions for LDA or QDA
- calculate confusion matrix for training data using the discriminant function
- calculate estimated classification error rates, APER and/or  $AER_{\text{hat}}$

## New data

- observe new data  $\mathbf{x}_0$
- classify  $\mathbf{x}_0$  as member of population 1 or 2 using the calculated discriminant functions for the training data
- calculate posterior probabilities using
  - $P(\pi_1|\mathbf{x}_0) \propto p_1 f(\mathbf{x}_0|\pi_1)$  (posterior<sub>1</sub>  $\propto$  prior<sub>1</sub>·likelihood<sub>1</sub>)
  - $P(\pi_2|\mathbf{x}_0) \propto p_2 f(\mathbf{x}_0|\pi_2)$  (posterior<sub>2</sub>  $\propto$  prior<sub>2</sub>·likelihood<sub>2</sub>)
  - $P(\pi_1|\mathbf{x}_0) + P(\pi_2|\mathbf{x}_0) = 1$  (normalization to actual probabilities)